

Assignment – 8

Question – 1

Code:

```
1/*
2Assign a pointer to an integer variable, character variable and a floating-point variable. Let each of these variables be initialised with a suitable value while declaring. Print
the address of the variables, pointer value and value stored at pointer (using * operator) to verify pointer concept. Further using the '++' operator increment each pointer and
print the pointer value and value stored at the pointers.
3*/
4
5#include <stdio.h>
6
7int main() {
8    int i = 3, *ip;
9    char c = 'A', *cp;
10   float f = 5.25, *fp;
11
12   ip = &i;
13   cp = &c;
14   fp = &f;
15
16   printf("Before Incrementing:\n");
17   printf("Addresses of variables are: %u, %u, %u.\n", &i, &c, &f);
18   printf("Pointer values of variables are: %u, %u, %u.\n", ip, cp, fp);
19   printf("Values stored in pointer are: %d, %c, %f.\n", *ip, *cp, *fp);
20
21   ip++; cp++; fp++;
22   printf("\nAfter Incrementing:\n");
23   printf("Addresses of variables are: %u, %u, %u.\n", &i, &c, &f);
24   printf("Pointer values of variables are: %u, %u, %u.\n", ip, cp, fp);
25   printf("Values stored in pointer are: %d, %c, %f.\n", *ip, *cp, *fp);
26 }
```

Output:

```
students@cslabiadn-in-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_8$ ./q1
Before Incrementing:
Addresses of variables are: 460970152, 460970151, 460970156.
Pointer values of variables are: 460970152, 460970151, 460970156.
Values stored in pointer are: 3, A, 5.250000.

After Incrementing:
Addresses of variables are: 460970152, 460970151, 460970156.
Pointer values of variables are: 460970156, 460970152, 460970160.
Values stored in pointer are: 1084751872, , 0.000000.
```

Question – 2

Code:

```
1/*
2Let main() function have a variable "num". Write 2 functions. First one to which "num" being passed by value. For the second function "num" to be passed by reference. Add a
constant inside the functions to the number passed as argument. Print the value of the resulting number inside the function before and after adding constant. Also print "num" in
main() function after the function calls are done. Observe the output.
3*/
4
5#include <stdio.h>
6void byValue(int n) {
7    printf("\nCalling by Value:\n");
8    printf("Before Incrementing, num = %d.\n", n);
9    n++;
10   printf("After Incrementing, num = %d.\n", n);
11}
12
13void byRef(int *n) {
14    printf("\nCalling by Reference:\n");
15    printf("Before Incrementing, num = %d.\n", *n);
16    (*n)++;
17    printf("After Incrementing, num = %d.\n", *n);
18}
19
20int main() {
21    int num;
22    printf("Enter a number: ");
23    scanf("%d", &num);
24    byValue(num);
25    printf("After Calling Function, num = %d.\n", num);
26    byRef(&num);
27    printf("After Calling Function, num = %d.\n", num);
28 }
```

Output:

```
students@cslabiadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_08 ./q2
Enter a number: 6

Calling by Value:
Before Incrementing, num = 6.
After Incrementing, num = 7.
After Calling Function, num = 6.

Calling by Reference:
Before Incrementing, num = 6.
After Incrementing, num = 7.
After Calling Function, num = 7.
```

Question – 3

Code:

```
1/*
2Assume there is a 10 element array declared in main. Write a function "readArray()" to which reference to the starting address of the array and number of elements need to be passed
3as arguments. Inside the function readArray() read the necessary number of elements from user. From the main() function, print the elements of the array after the readArray() is
4executed.
5*/
6#include <stdio.h>
7void readArray(int *b, int n) {
8    for(int i = 0; i < n; i++) printf("%d ", *(b+i));
9    printf("\n");
10}
11int main() {
12    int a[10];
13    printf("Enter 10 integers: ");
14    for(int i = 0; i < 10; i++) scanf("%d", &a[i]);
15    printf("The 10 element array is: ");
16    readArray(a, 10);
17}
```

Output:

```
students@cslabiadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_08 ./q3
Enter 10 integers: 1 3 5 7 9 11 34 56 88 90
The 10 element array is: 1 3 5 7 9 11 34 56 88 90
```

Question – 4

Code:

```
1/*
2Add a function "printArray()" to the program written to question 3 above that takes responsibility of printing the elements of the array.
3*/
4
5#include <stdio.h>
6void printArray(int x) {
7    printf("%d ", x);
8}
9
10void readArray(int *b, int n) {
11    for(int i = 0; i < n; i++) printArray(*(b+i));
12    printf("\n");
13}
14
15int main() {
16    int a[10];
17    printf("Enter 10 integers: ");
18    for(int i = 0; i < 10; i++) scanf("%d", &a[i]);
19    printf("The 10 element array is: ");
20    readArray(a, 10);
21}
```

Output:

```
students@cslabiadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_08 ./q4
Enter 10 integers: 1 2 3 5 6 7 9 0 23 56
The 10 element array is: 1 2 3 5 6 7 9 0 23 56
```

Question – 5

Code:

```
1/*
2From questions 3 and 4 you have 2 functions: readArray() and printArray(). Same functions need to be used for this question also. In main() function asks the user to enter the
3index of the array at which the value to be modified. Pass this index and reference to the array to third function scaleElement() that adds a constant to the mentioned element of
4the array. Before calling scaleElement() use printArray() to print the array. After calling scaleElement(), use printArray() again to print the array in main() function.
5*/
6#include <stdio.h>
7void printArray(int x) {
8    printf("%d ", x);
9}
10void readArray(int *b, int n) {
11    for(int i = 0; i < n; i++) printArray(*(b+i));
12    printf("\n");
13}
14void scaleElement(int *b, int n, int index, int cst) {
15    *(b+index) += cst;
16    for(int i = 0; i < n; i++) printArray(*(b+i));
17    printf("\n");
18}
19}
20
21int main() {
22    int a[10], index, cst;
23    printf("Enter 10 integers: ");
24    for(int i = 0; i < 10; i++) scanf("%d", &a[i]);
25    printf("The 10 element array is: ");
26    readArray(a, 10);
27
28    printf("Enter the index of element to scale: ");
29    scanf("%d", &index);
30    printf("Enter the constant to scale: ");
31    scanf("%d", &cst);
32    printf("The new 10 element array is: ");
33    scaleElement(a, 10, index, cst);
34}
```

Output:

```
students@cslabiadmin-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_8$ ./q5
Enter 10 integers: 1 2 3 4 5 6 7 8 9 10
The 10 element array is: 1 2 3 4 5 6 7 8 9 10
Enter the index of element to scale: 5
Enter the constant to scale: 5
The new 10 element array is: 1 2 3 4 5 11 7 8 9 10
```